

Cerebral blood flow monitoring using a deep learning implementation of the two-layer diffuse correlation spectroscopy analytical model with a 512×512 SPAD array

ABSTRACT. Significance: Multilayer (two- and three-layer) diffuse correlation spectroscopy (DCS) models improve cerebral blood flow index (CBFi) measurement sensitivity and mitigate interference from extracerebral tissues. However, their reliance on multiple predefined parameters (e.g., layer thickness and optical properties) and high computational load limit their feasibility for real-time bedside monitoring.

Aim: We aim to develop a fast, accurate DCS data processing method based on the two-layer DCS analytical model, enabling real-time cerebral perfusion monitoring with enhanced brain sensitivity.

Approach: We employed deep learning (DL) to accelerate DCS data processing. Unlike previous DCS networks trained on single-layer models, our network learns from the two-layer DCS analytical model, capturing extracerebral versus cerebral dynamics. Realistic noise was estimated from subject-specific baseline measurements using a 512×512 SPAD array at a large source-detector separation (35 mm). The model was evaluated on test datasets simulated with a four-layer slab head model via Monte Carlo (MC) methods and compared against conventional single-exponential fitting and the two-layer analytical fitting. Two in vivo physiological response tests were also conducted to assess the real-world performance.

Results: The proposed method bypasses traditional curve-fitting and achieves real-time monitoring of CBF changes at 35 mm separation for the first time with a DL approach. Validation on MC simulations shows superior accuracy in relative CBFi estimation and significantly enhanced CBFi sensitivity. Although the two-layer analytical fitting offers optimal performance, it depends on strict assumptions and preconditions, and its computational complexity makes it time-consuming and unsuitable for real-time monitoring applications.

In addition, our method minimizes the influence of superficial blood flow and is 750-fold faster than single-exponential fitting in a realistic scenario. In vivo tests further validated the method's ability to support real-time cerebral perfusion monitoring and pulsatile waveform recovery.

Conclusions: This study demonstrates that integrating DL with the two-layer DCS analytical model enables accurate, real-time cerebral perfusion monitoring without sacrificing depth sensitivity. The proposed method enhances CBFi sensitivity and recovery accuracy, supporting future deployment in bedside neuro-monitoring applications.

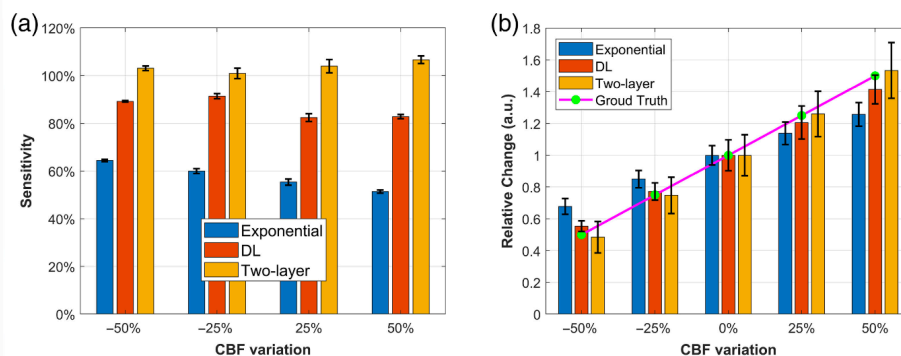


Fig. 8 (a) and (b) Bar charts showing the recovered brain sensitivity and relative changes in response to CBFi variations are presented for single-exponential fitting, the DL model, and the two-layer analytical fitting. Note: We applied nonparametric bootstrapping to quantify the uncertainty in the estimated mean values of the sensitivity data as the original standard deviations were relatively large, making it difficult to compare results across different methods. Specifically, we randomly resampled 1000 data points from the original sensitivity data and calculated their mean. This process was repeated 1000 times, resulting in a distribution of bootstrap means. The error bars in panel (b) represent the 25th and 75th percentiles of the recovered data, providing an interquartile range-based measure of variability.

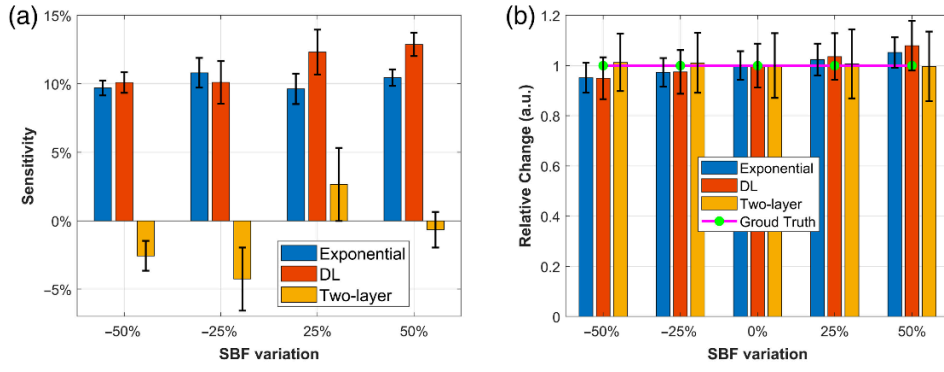
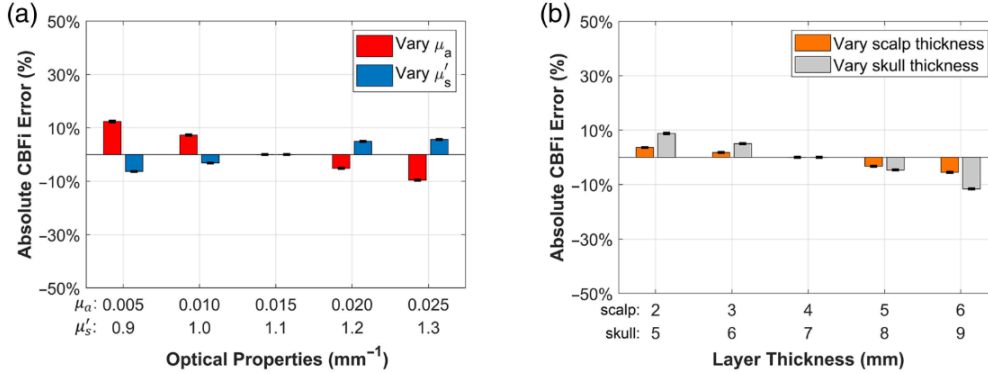


Fig. 9 (a) and (b) Bar charts showing the recovered brain sensitivity and relative change in response to SBFi variation for single-exponential fitting, DL model, and the two-layer fitting. The error bars in panel (a) were calculated using the bootstrap distribution of mean values, following the same procedure used for the CBF sensitivity analysis. The error bars represent the 25th and 75th percentiles of the recovered data.

DL model



Two-layer analytical model

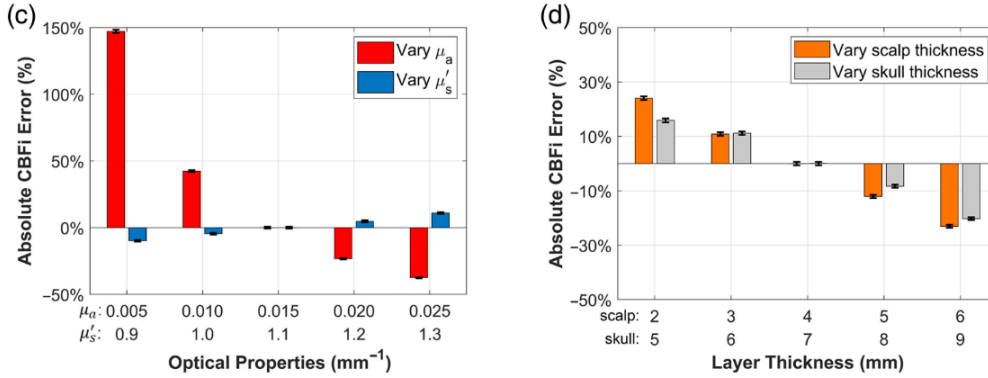
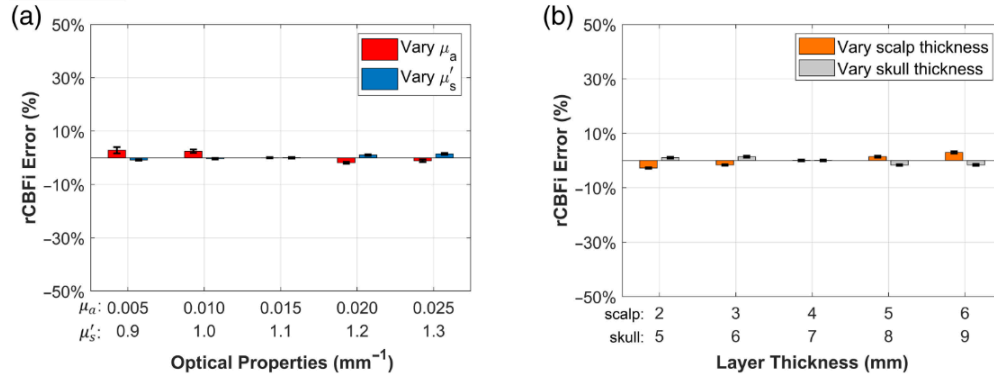


Fig. 10 Comparison of absolute CBFi recovery between the DL method and the two-layer analytical model fitting. Bootstrapping was applied to estimate the mean CBFi errors from 1000 data points, following the same procedure described in the previous section. CBFi errors were calculated using Eq. (13). When varying one parameter, all other parameters were fixed at their baseline values ($\mu_a = 0.015$, $\mu'_s = 1.1$, $L_{\text{scalp}} = 4$ mm, and $L_{\text{skull}} = 7$ mm). These baseline values were also assumed in the two-layer analytical model fitting for consistency.

DL model



Two-layer analytical model

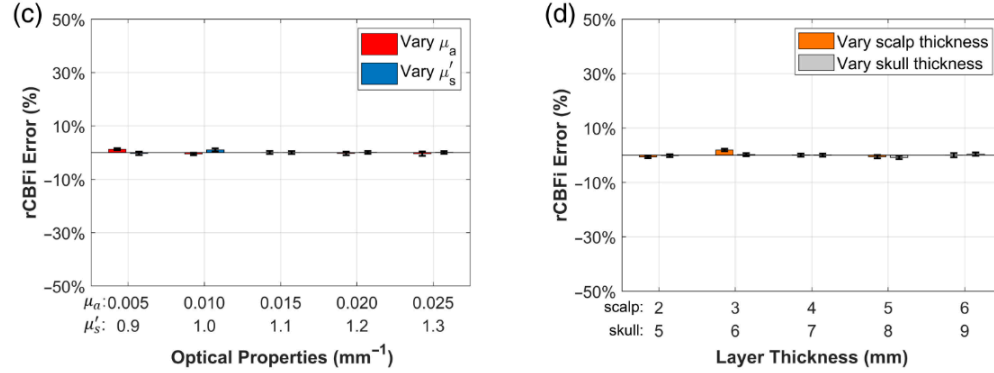


Fig. 11 Comparison of rCBFi recovery between the DL method and the two-layer analytical model fitting. The same strategy used in Fig. 10 for absolute CBFi comparison was applied here. rCBFi was calculated based on $\pm 50\%$ perturbations relative to the baseline condition. The displayed rCBFi errors were computed using Eq. (13), with the errors from the two perturbation levels averaged to remove bias related to the direction of variation.

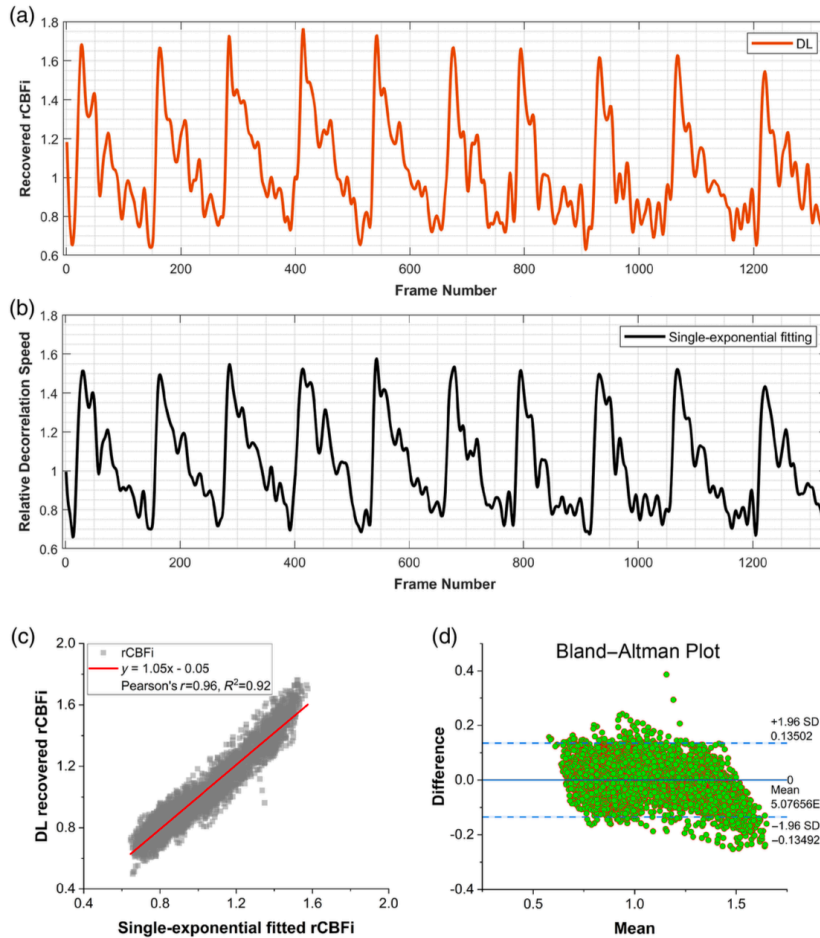


Fig. 12 Recovered baseline cerebral perfusion waveform (30 min before lunch). (a) and (b) The rCBFi waveform was filtered using a third-order Butterworth low-pass filter (cutoff frequency = 0.15, normalized) with zero-phase filtering (filtfilt) to reduce high-frequency noise. The mean value of recovered CBFi by DL and the mean value of recovered decorrelation speed by single-exponential fitting were taken as the baselines to calculate the relative cerebral perfusion changes, respectively. (c) Pearson's correlation analysis and (d) Bland-Altman analysis between the DL model and the single-exponential fitting recovered the CBF waveform.